

Null Values



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The Two Types of Data

Not-nullable
(Can never be null)

Nullable
(Might be null)



Being explicit about this
helps code robustness



Overview



Null values

- How to use nullable and not-nullable types
 - Different techniques for value and reference types
- Null vs. empty for strings
- Enable nullable reference types on an existing project



Demo



Making structs nullable

- Delivery times app
- Will feature nullable and not-nullable structs





Nullability for Reference Types



Demo



Delivery times app – but with classes

- Will show similar nullability syntax
- But differences in techniques required for classes



**The following demo
presumes nullable
reference types are enabled**



Nullable structs

`MyStruct?` and `MyStruct`
are different types

(`MyStruct?`, under the hood
becomes
`System.Nullable<MyStruct>`)

vs

Nullable classes

`MyClass?` and `MyClass`
are the same type

All classes are intrinsically
nullable under the hood

Declaring `MyClass` tells the compiler you
don't want to put `null` in this instance...

... and the compiler should warn you if it
thinks your code might store `null` in it



Nulls and Classes: If Null-Checking is Not Enabled



Demo

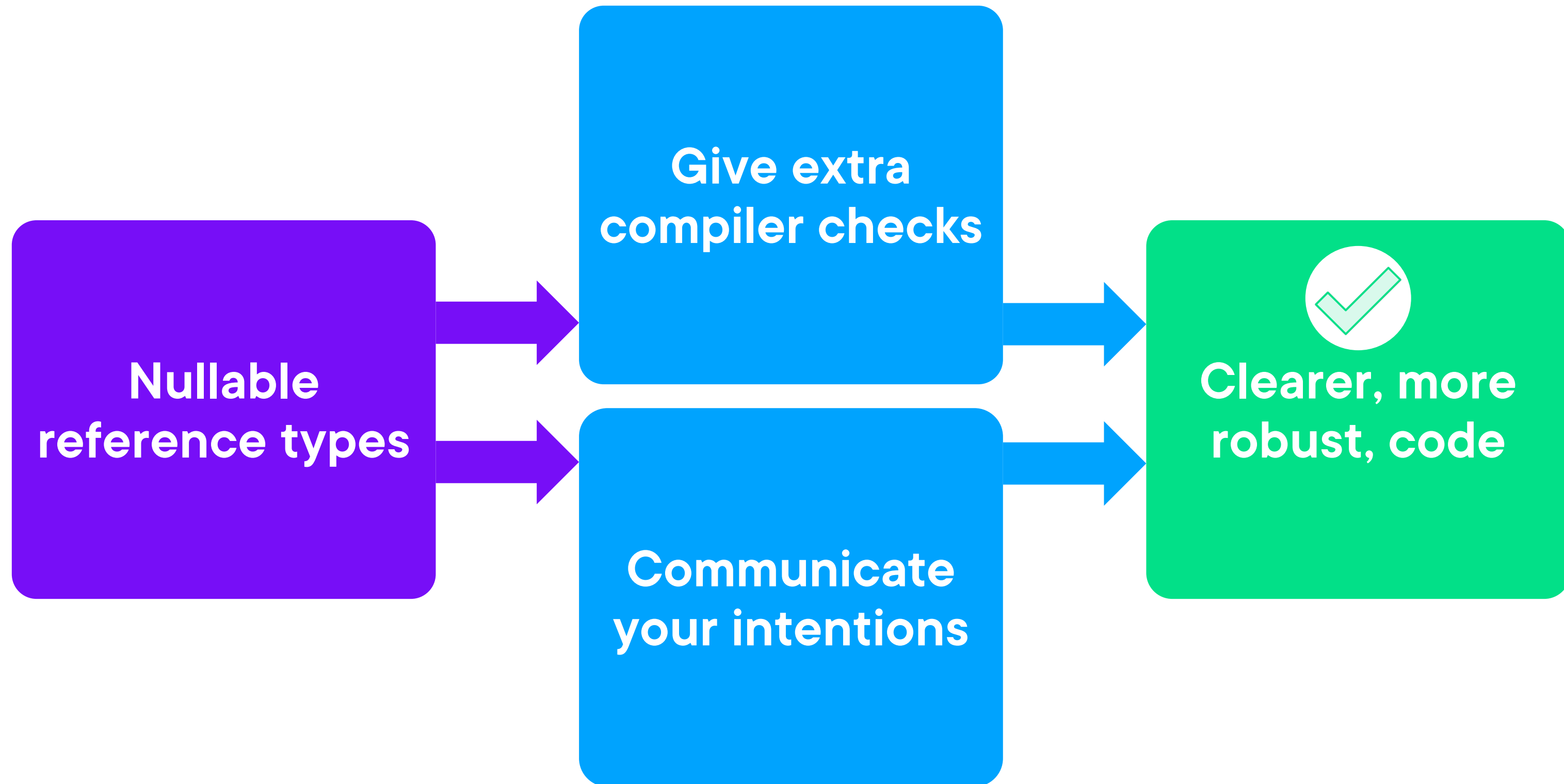


Delivery times project (again)

- This time, without nullable reference types enabled



Enabling Nullable Reference Types





Null and Empty Strings



Demo



New demo

- Features a string that might lack a usable value
- Shows nullability is more complex for strings



String with No Data

```
string? value = null;
```

```
string value = "";
```

```
string value = "  ";
```

May need to prepare for all these cases

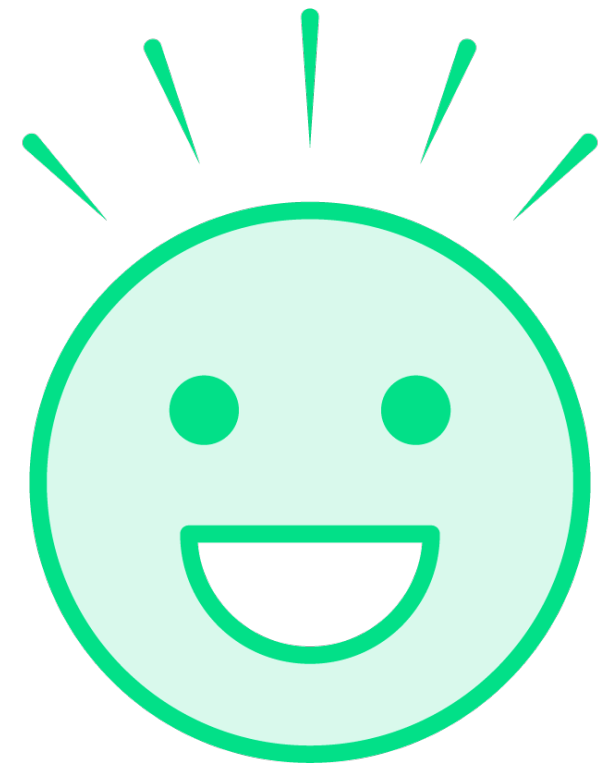
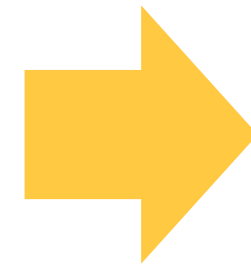
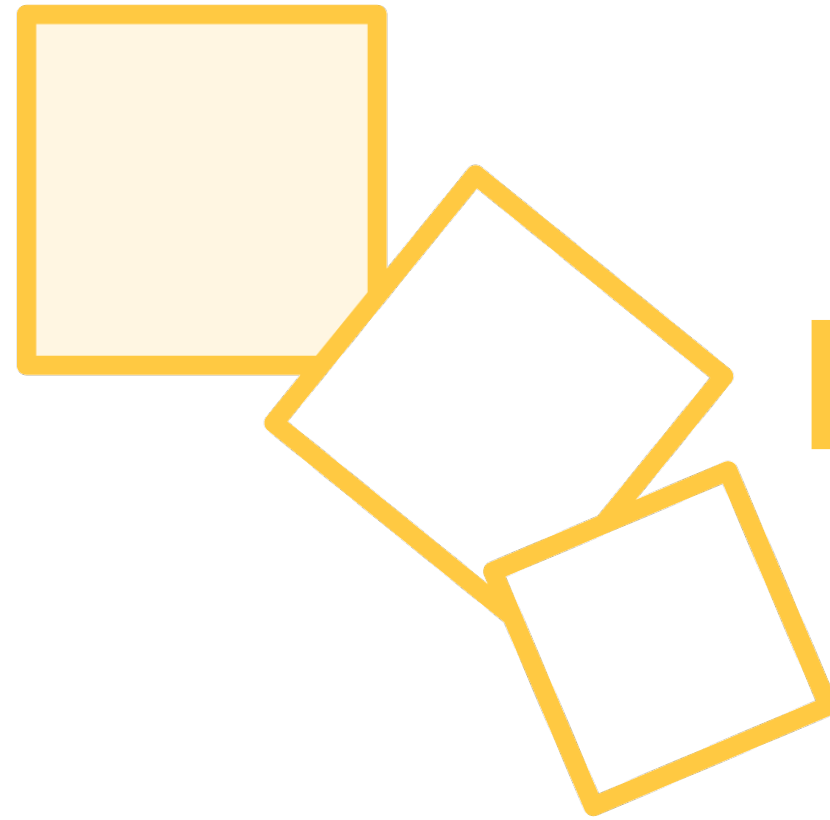
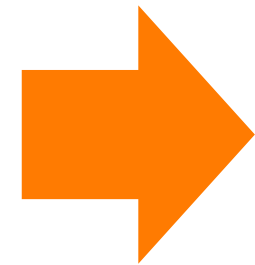
What significance do these values have?

Whitespace might or might not count as data – it's situation-dependant

Example: Preserve whitespace in an XML file



Strings



**Think what
null/whitespace
means**

**Transform to single
form if appropriate**

Get fewer bugs





Enabling Nullable Reference Types



Demo



Small library app

- Library of utility functions
- Written without nullable reference type checking
- Will enable nullable reference types
- App has no comments to clarify author's intentions (!)



Summary



Null values

- Use ? to declare an instance as nullable
- For structs, that gives a different type
 - Value property to get value
- For classes, depends on enabling nullable reference types
 - Same type, just compiler heuristic checking

Strings

- Think about empty and whitespace strings as well as null

Enable nullable reference types

- Decide which instances are nullable

